AKANSHA SUBEDI

STRUCTURAL ENGINEER

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STATEMENT

Experienced Structural Engineer specializing in steel and reinforced concrete design for complex buildings and bridge infrastructure. Over 5 years of proven expertise in advanced linear and non-linear structural analysis, working across multiple international standards (ACI, Eurocode, IS, Thai codes) to deliver optimized solutions. Skilled in Performance-Based Design methodologies with a track record of providing innovative, safety-critical engineering solutions for diverse infrastructure projects

WORK EXPERIENCES

Structural Design Engineer *(Freelance – Thailand & Nepal)* ***Jan 2024 – Present***

- Performed structural analysis and design according to AISC, ASCE & Thai code (DPT 1311-50) and provided detailed drawings and calculation reports of numerous Steel framed residential buildings, commercial buildings, car park structure and warehouse with truss roofing in Koh Samui, Thailand
- Carried out analysis and design in ETABS and manual calculation for the foundations, base plate and anchor bolts of steel structure for indoor sports facility with truss roof and wall climbing structure in Koh Samui
- Carried out analysis and design of Reinforced Concrete buildings in ETABS, SAP2000 and SAFE for foundation, according to Nepali and Indian codes (IS 456 and NBC (105:1994)) of more than 10 residential, commercial and community buildings in Nepal
- CSiBridge -ET (a FEM-based software for bridge analysis and design) – QA testing and preparation of documentation and user manuals

Structural Engineer (*Computers and Structures Inc. (CSI) Bangkok)* ***June 2023 – Dec 2023***

- Performed manual calculations for analysis and design of concrete bridge components (PSC Girders, Bent Caps, Columns etc.) and verified the results from the software for QA testing of CSiBridge-ET software
- Performed detailed design of concrete bridge components and developed mind maps and algorithms for design steps by following various codes (AASTHO LRFD, Caltrans, Eurocode)

Civil Engineer *(TEAM Consultants Pvt. Ltd., Nepal)* ***April 2017 – Oct 2019***

- Performed linear and non-linear, static and dynamic analysis, calculations, and seismic design of more than 15 building structures
- Provided detailed structural drawings, calculation reports, cost estimations and BOQ
  - Low-rise to mid-rise RC framed buildings of residential and commercial complexes
  - Core shear wall and outer RC framed structure of a commercial complex
  - Extension of a hospital building with expansion joint between new and old structures

EDUCATION

Master in Structural Engineering ***2020 – 2022***
Asian Institute of Technology, Thailand (GPA: 3.44)

Bachelor in Civil Engineering ***2011 – 2015***
Tribhuvan University, Nepal

TECHNICAL SKILLS

ETABS	AutoCAD	CSiBridge	MATLAB
SAP2000	PERFORM 3D	REVIT	Python
STAAD.Pro	SAFE	MS Excel	VecTor FormWorks

OTHER EXPERIENCE

Thesis in Master in Structural Engineering ***2021-2022***

Topic: "The Effect of Performance-Based Wind Design on Seismic Design of a Tall Building in High Seismic and High Wind Zone"

Thesis Advisor: Dr. Naveed Anwar

- 93 Storied building: RCC framed core wall and BRB provided for lateral force resistance
- Studied the effect of considering non-linearity in wind design on seismic ductile design of the building through performance-based approach

Student Research Assistant *(Asian Institute of Technology)* ***Jan 2021– April 2022***

- Assisted in writing a book titled "Strut and Tie Method"
- Conducted research on Advance Concrete Structures and Tall Buildings, and developed academic articles
- Conducted testing and documentation of software for CSi Inc. (CSi Detail & CSi Designer)

Intern Engineer *(HELVETAS Swiss Inter-cooperation, Nepal)* ***June 2016– March 2017***

- Performed complete survey of water supply system from the water source to households and designed infrastructures such as intake, reservoirs, and pipelines

Intern Engineer *(Hydro-Consult Engineering Ltd., Nepal)* ***April 2016– May 2016***

- Supported structural assessments of intake structures, spillways, and powerhouse buildings
- Prepared technical drawings and FEM models under supervision of senior engineers

LICENSE AND CERTIFICATE

Professional Engineer Skill Assessment *(Engineers Australia)* *August 2024*

Registered Engineer *(Nepal Engineering Council)* *June 2016*

PUBLICATIONS

- "Moment Curvature Curve: Unlocking the Hidden Treasures"
- "Basic Concepts for Considering Soil-Structure Interaction"

LANGUAGES

English (Proficient)- **PTE (A) - 77** **Nepali** (Native)